Adrenergic receptor blockers (Sympatholytic drugs)

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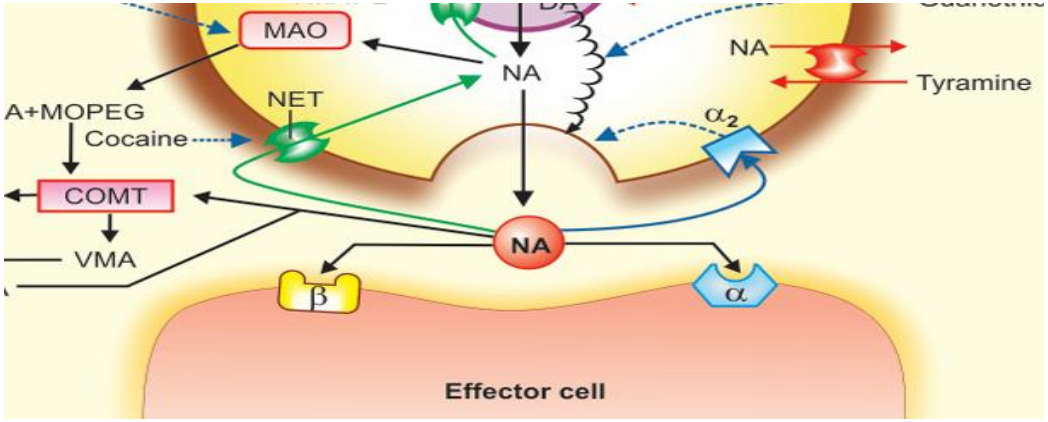
The adrenergic receptor blockers (Sympatholytic agents)

Sympatholytic agents are drugs which inhibit the action of catecholamine on adrenergic receptors.

These drugs act by either reversibly or irreversibly binding to the adrenoceptors, thus preventing their activation by endogenous catecholamines.

They are divided into

- α-receptor blockers
- β-receptor blockers



Adrenergic antagonists have important roles in clinical medicine, primarily to treat diseases associated with the cardiovascular system

α-adrenergic receptor blockers

Non- Selective α-receptor blockers

They block both α1and α2 adrenergic receptors, which includes. both **phenoxybenzamine** and **phentolamine**. Drugs that block α adrenoceptors profoundly affect blood pressure.

A. Phenoxybenzamine: It's the prototype drug. It blocks irreversibly and noncompetitively both α1 and α2 receptors by binding with a covalent bond. Therefore, it has long duration of actions (about 24 hours).

By blocking α1 receptors it cause vasodilatation (decrease blood pressure) which provokes a reflex tachycardia, and because of its ability to block presynaptic inhibitory α2 receptors in the heart results in more norepinephrine release, which stimulates β1-receptors on the heart (cause tachycardia), which contribute to an increased cardiac output.

Therefore, this drug is unable to maintain the lowered blood pressure in hypertension patients.

Therapeutic uses:

- Used in the treatment of pheochromocytoma, prior to surgical removal of the tumor or for chronic treatment of inoperable tumors
- Can be used in treatment of Raynaud disease and frostbite

Adverse effects:

- Postural hypotension
- Nasal stuffiness
- Myosis
- It may inhibit ejaculation
- It cause reflex tachycardia which is mediated by the baroreceptors (should not be used in patients with angina)

B. Phentolamine: It's a competitive inhibitor of α1 and α2 receptors its effect lasts for approximately 4 hours.

Selective α-receptor blockers

Selective α1-receptor blockers

Prazosin, terazosin, doxazosin, tamsulosin, and alfuzosin

They are selective competitive blockers of the α1 receptor, they are useful in the treatment of hypertension.

Prazosin has a short duration of action. The other drugs have longer durations of action, so they can be given once daily.

Mechanism of action: they decrease peripheral vascular resistance and lower blood pressure by causing relaxation of both arterial and venous smooth muscle. They cause less tachycardia and minimal changes in cardiac output.

Tamsulosin and **Alfuzosin:** more selective for α1A receptors in the prostate and bladder and less effect on blood pressure because it is less selective for α1B receptors found in the blood vessels. It improves urination in patients with benign prostatic hyperplasia (BPH). They are better to be prescribed for patients with BPH and HT

Adverse effects:

- Headach, dizziness and drowsiness
- Orthostatic hypotension with less reflex tachycardia
- Nasal congestion

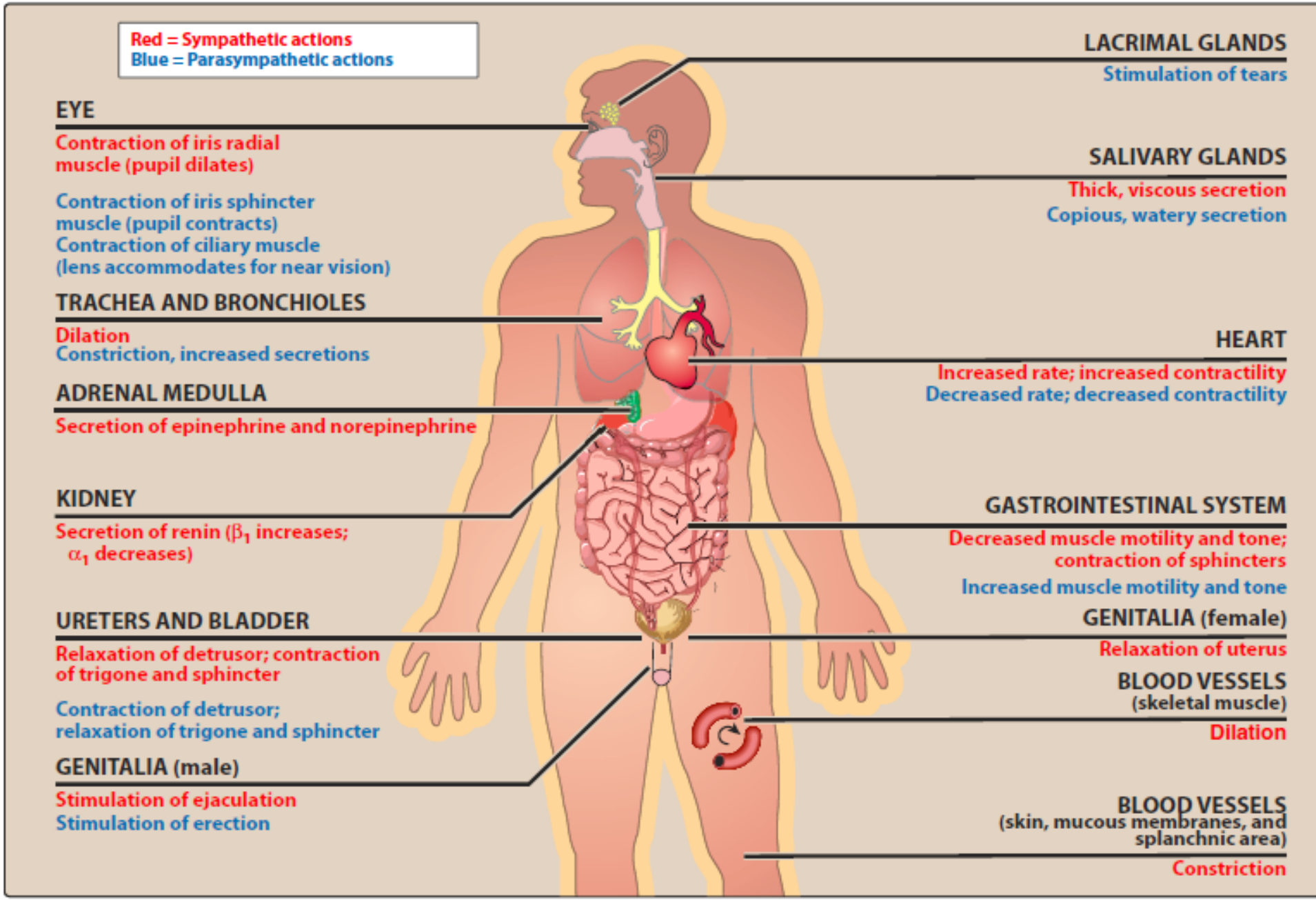


Figure 3.3
Actions of sympathetic and parasympathetic nervous systems on effector organs.

Selective α2-receptor blockers

Yohimbine: it has been used for treatment of erectile dysfunction, now a days not recommended. It is contraindicated in cardiovascular disease, psychiatric conditions, and renal dysfunction because it may worsen these conditions.

β-adrenergic receptor blockers

Non-selective β blockers act on both β1 and β2 receptors, whereas cardio-selective β receptor antagonists primarily block β1 receptors and all of them are competitive antagonists.

[There are no clinically useful β2 antagonists]

- **Non-selective β-receptor blockers such as:**

Propranolol, Sotalol, Timolol, Nadolol, Pindolol

Propranolol is the prototype, it blocks both β1 and β2 receptors with equal affinity.

- **Selective β-receptor blockers**

Cardio-selective β blockers (block β1 receptors), such as:

Metoprolol, Atenolol, Esmolol

They preferentially block the β1 receptors minimize the unwanted bronchoconstriction (β2 effect) seen with propranolol use in asthma patients. (this cardioselectivity will be lost at high doses.)

Esmolol has a very short half-life. It is only available intravenously and is used to control blood pressure and heart rate during surgery or diagnostic procedures

Nebivolol: In addition to its cardioselectivey, it releases nitric oxide from endothelial cells and causes vasodilation.

- **β blockers with additional alpha-blocking or vasodilatory properties.**

Labetalol, Carvedilol, Nebivolol

Carvedilol also decreases lipid peroxidation and vascular wall thickening, effects that have benefit in heart failure (anti-inflammatory effects), anti-apoptotic and anti-oxidant properties. For this reason, it's also called a multiple-action neurohumoral antagonist.

- **β blockers with intrinsic sympathomimetic activity (Antagonists with partial agonist activity) such as:**

Acebutolol and pindolol

These drugs are not pure antagonists but also have the ability to weakly stimulate both β1and β2 receptors.

β-blockers with ISA are effective in hypertensive patients with moderate bradycardia (not used for stable angina or arrhythmias)

Pharmacokinetics: After oral administration, propranolol is almost completely absorbed. It is subject to first-pass effect, and only about 25% of an administered dose reaches the circulation. It has large volume of distribution and readily crosses the blood–brain barrier due to its high lipophilicity.

Propranolol is extensively metabolized in the liver, and are contraindicated in liver disease and its metabolism largely depends on hepatic blood flow.

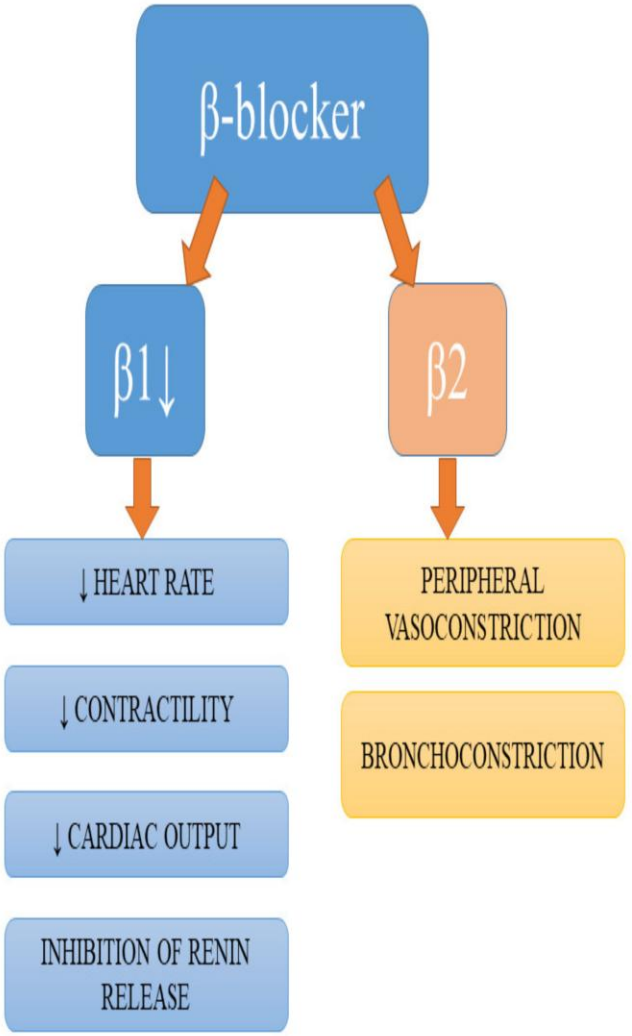
Atenolol, Nadolol and Sotalol are water soluble β-blockers mainly excreted unchanged by the kidneys therefore their dose should be adjusted in patients with kidney impairment.

Pharmacological actions

Effect on the heart:

- -ve inotropic effects (decrease cardiac contractility)
- -ve chronotropic effects (decrease heart rate)
- It decrease cardiac output and, leading to reduction in blood pressure.
- Depress SA and AV nodal activity they are effective in attenuating supraventricular cardiac arrhythmias
- During exercise or stress when the sympathetic nervous system is activated, β-blockers reduce the expected increase in heart rate. Cardiac output, workload, and oxygen consumption are decreased by blockade of β1 receptors, and these effects are useful in the treatment of angina.

Peripheral vasoconstriction: Nonselective blockade of β receptors prevents β2-mediated vasodilation in skeletal muscles, increasing peripheral vascular resistance.



Effects on bronchial muscles: Normally β2 stimulation causes bronchodilation, so blocking β2 receptors causes contraction of the bronchiolar smooth muscle, this can precipitate an exacerbation in patients with chronic obstructive pulmonary disease (COPD) or asthma. Therefore, β-blockers, particularly, nonselective ones, are contraindicated in patients with COPD or asthma.

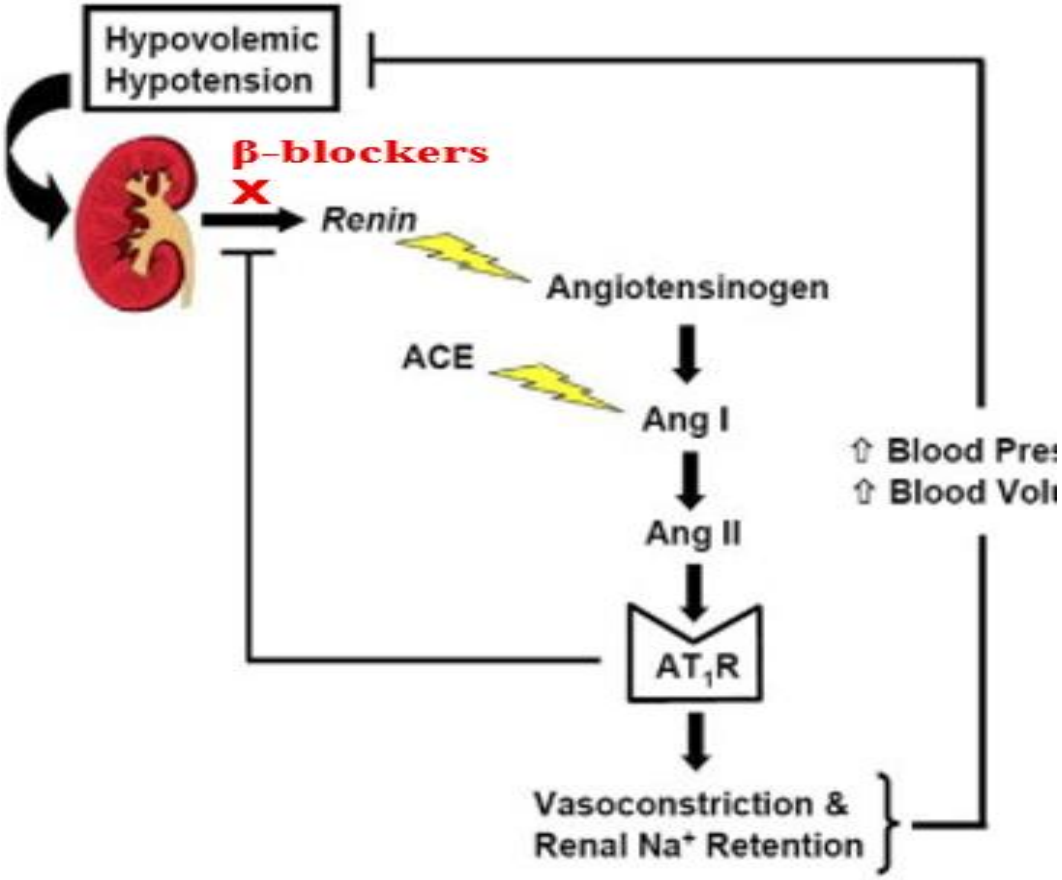
Disturbances in glucose metabolism: β blockade leads to decreased glycogenolysis and decreased glucagon secretion.

Therefore, if propranolol is given to a diabetic patient receiving insulin, careful monitoring of blood glucose is essential, because pronounced hypoglycemia may occur after insulin injection.

β-blockers also mask the symptoms of hypoglycemia.

Effects on the kidneys:

- β blockers decrease renin release.



Therapeutic uses:

Hypertension: Propranolol does not reduce blood pressure in people with normal blood pressure. It lowers blood pressure in hypertension by several different mechanisms of action.

- Decreased cardiac output is the primary mechanism
- Inhibition of renin release from the kidney
- Decrease in total peripheral resistance with long-term use
- Decreased sympathetic outflow from the CNS

Angina pectoris: Propranolol decreases the oxygen requirement of heart muscle. thus, its useful in the chronic management of stable angina

Myocardial infarction: Propranolol and other β-blockers have a protective effect on the myocardium. Also reduces infarct size and hastens recovery.

Migraine: Propranolol is effective in reducing migraine episodes when used prophylactically, due to its lipophilic nature that allows it to penetrate the CNS.

Hyperthyroidism: The extensive sympathetic activation that occurs in hyperthyroidism can be effectively reduced by propranolol and other β-blockers. Also inhibit conversion of T4 to T3.

Anxiety: The drug used is the lipid-soluble β blocker propranolol.

Timolol: reduces the production of aqueous humor by the ciliary body in the eye and diminishing intraocular pressure. It is used topically in the treatment of glaucoma.

Nadolol: has a very long duration of action used for treatment of angina and hypertension.

Adverse effects:

- Tiredness, fatigue, and weakness
- Cold hands and Raynaud's phenomena (less common in drugs with intrinsic sympathomimetic activity)
- Bronchoconstriction.
- Arrhythmias: Treatment with β-blockers must never be stopped abruptly because of the risk of precipitating cardiac arrhythmias, which may be severe.

Bradycardia (less common in β blockers with ntrinsic sympathomimetic activity) & heart attack.

- β blockers inhibit glucose production (by glycogenolysis) induced by adrenaline (sympathetic nervous stimulation), causing hypoglycemia.

- Disturbance of lipid metabolism, lipases in fat cells are activated mainly by β2 and β3 receptor stimulation. Patients administered nonselective β-blockers have increased low density lipoprotein (LDL) increased triglycerides and decreased HDL
- CNS effects: including headache, dizziness, vivid dreams, nightmares and hallucination specially with lipid soluble β-blockers.

Contraindications

- Asthma
- Severe heart failure
- Type I DM
- Hyperlipidemia

Drug interactions:

β blockers should not be given to patients receiving drugs which have negative ionotropic effects or cause bradycardia, because of the incidence of heart block/cardiac arrest.

Some drugs inhibit, the metabolism of propranolol, such as cimetidine, fluoxetine and potentiate its antihypertensive effect